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ml & genai research engineer · 12 from-scratch pytorch projects across llms, diffusion, multimodal, and agentic systems
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SUMMARY

ML & GenAI Research Engineer with 12 from-scratch PyTorch projects across LLMs, latent diffusion, multimodal, and agentic ML systems. Headline results: 78% peak-memory reduction on LLaMA-3 pretraining (92 GB to 20 GB), 0.0947 training loss on Stable Diffusion 1.x from scratch (860M UNet, 2x RTX 5090), and a 15-phase multi-agent ML research platform with 878 passing tests. Specialized in GPU memory engineering, paper-faithful reproductions, and provider-agnostic agentic orchestration.

CORE STACK

python 3.12 · pytorch 2.x · torch.compile · bf16 · flash attention 2 · gradient checkpoint · chunked ce · ddp · nccl · safetensors · wandb
transformers · mla · moe · mtp · gated delta net · ddpm · ddim · vae · clip · siglip · pydantic v2 · multi-agent · vector + graph memory

SELECTED PROJECTS

- deepseek-v3-lite** · llm Faithful from-scratch reimplementation of DeepSeek-V3 — multi-head latent attention with decoupled RoPE, aux-loss-free MoE routing, multi-token prediction, and MLA absorption at inference. Architecture, training pipeline, microbench, and step-time benchmarks complete; Chinchilla-optimal 422M / 8.4B-token run designed for a single A100 80GB. Companion 643-line MLA deep-dive.
- stable diffusion 1.x from scratch** · diffusion Trained a full SD 1.x latent diffusion model from random init on 2x RTX 5090 (Blackwell). 860M-param UNet across a 7-phase curriculum on 1.3M+ images (LAION-aesthetic → high-aesthetic LAION → DiffusionDB/JourneyDB → VGGFace2 → COCO → consolidation); best loss 0.0947 at epoch 16. 42-epoch checkpoint publicly released on Hugging Face. Min-SNR loss, GPU-resident EMA, channels_last, DDP/NCCL.
- vision language model** · multimodal PaliGemma-inspired VLM built from scratch — SigLIP-style vision encoder with sinusoidal patch embeddings, GQA language decoder with RoPE and GeGLU, trainable linear projector. Zero pretrained weights; trained end-to-end on COCO 2014 captions on a single P100.
- autonomous ml research engineer** · agentic platform 15-phase multi-agent platform that turns an arXiv paper into evaluated experiments — paper analysis, repo analysis, experiment planning, code patches, training runs, statistical evaluation, autonomous looping, end-to-end research reports. 23 agents, 61 typed tools, 186 Pydantic models, 878 passing tests. Provider-agnostic LLM layer with per-agent model routing.
- llama-3-lite** · llm From-scratch LLaMA-3-style transformer (515M params) engineered to fit on a single A100 80GB. 78% peak-memory reduction (92 GB to 20 GB) via gradient checkpointing, chunked cross-entropy (logits 50 GB to 0.3 GB), and disk-backed token caching (RAM 112 GB to 1 MB) — enabling 2x batch-size headroom. GQA, RoPE (theta=500K), fused SwiGLU, RMSNorm, FA2, BF16.
- fusionllm** · llm research Hybrid LLM pre-training framework fusing MLA, Gated Delta Net (linear-attention sequence mixer), DeepSeek-style MoE, and multi-token prediction into a 24-layer stack. 415.6M active / 868.6M stored params. Dual optimizer (NorMuon + CautiousAdamW), WSD + mu-P LR schedule, logit softcap.

EXPERIENCE

- ml engineering portfolio** · self-directed, full-time · nov 2022 – present Designed, implemented, and trained 12 end-to-end systems in raw PyTorch — no high-level wrappers where the model is the subject. Coverage: LLM pretraining (DeepSeek-V3 reproduction, LLaMA-3, FusionLLM hybrid), latent diffusion (SD 1.x from scratch on 1.3M+ images), multimodal (PaliGemma-style VLM on COCO), generative vision (CycleGAN, beta-VAE, DCGAN), video understanding (HRNet pose + ST-GCN), and an agentic ML research platform. Every project engineered to fit on a single A100 80GB, 2x RTX 5090, or smaller via BF16, FA2, gradient checkpointing, torch.compile, and chunked cross-entropy.

EDUCATION

- b.tech, civil engineering** · heritage institute of technology, kolkata · 2020 – 2024 Self-taught ML in parallel with coursework. Civil engineering gave a foundation in numerical methods, optimization, and linear algebra that maps cleanly onto modern ML.